Graphs — Direct proportion · Real-life graphs · Distance–time · Rates of change

You will need squared paper for the questions that say *draw*. ★ = challenge question.

A · DIRECT PROPORTION AND REAL-LIFE GRAPHS

1 The table shows the cost of apples at a market: 2 kg cost €5, 4 kg cost €10, 7 kg cost €17.50.

- **a** Are cost and mass in direct proportion? Explain how you know.
- **b** Draw a graph of cost against mass for 0 to 8 kg.
- **c** Use your graph to find the cost of 11 kg.
- **d** How many kilograms can you buy for €40?

> **a** Yes: the ratio cost ÷ mass is always €2.50 per kg (doubling the mass doubles the cost), so the graph is a straight line through the origin.
> **b** Straight line from $(0, 0)$ through $(4, 10)$ and $(8, 20)$.
> **c** Extend the line (or use the rate): Math input error. **d** $40 \div 2.50 = 16$ kg.

2 An electrician charges a call-out fee plus an hourly rate. Some charges: 2 hours → €70, 4 hours → €100, 5 hours → €115.

- **a** Draw the graph (time on the horizontal axis).
- **b** What is the hourly rate?
- **c** What is the call-out fee?
- **d** Are time and cost in direct proportion? Explain.

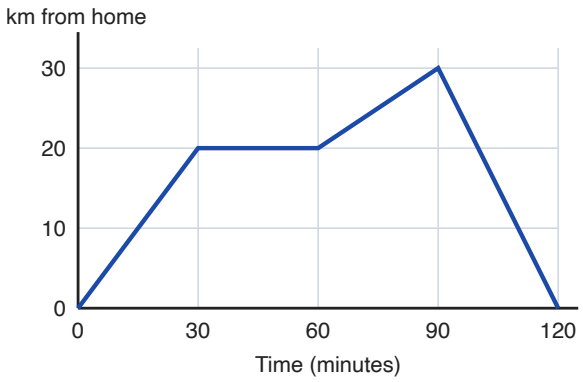
> **a** A straight line through $(2, 70)$, $(4, 100)$, $(5, 115)$.
> **b** From 2 to 4 hours the cost rises €30, so €15 per hour (the gradient).
> **c** Go back 2 hours from $(2, 70)$: Math input error — the cost at 0 hours.
> **d** No — the line does not pass through the origin (0 hours still costs €40), and doubling the time does not double the cost.

3 Four sketch graphs: **A** a straight line through the origin; **B** a curve getting steeper; **C** a straight line that starts above the origin; **D** a horizontal line. Which shows direct proportion? Explain why the others do not.

> Only **A**: direct proportion needs a straight line *through the origin*. B is not straight (the rate changes), C does not pass through the origin, D does not change at all.

B · DISTANCE–TIME GRAPHS

4 The graph shows Nadia's cycle ride from home.



a For how long did Nadia stop?

b How far from home was she at 90 minutes?

c Which part of the ride was fastest? How do you know?

d Work out her average speed on the way home, in km/h.

e What total distance did she cycle?

> **a** The flat part, 30 to 60 minutes: 30 minutes.
>
> **b** 30 km.
>
> **c** The last part (90–120 min): it is the steepest line — 30 km in 30 minutes.
>
> **d** 30 km in 0.5 h: $30 \div 0.5 = 60$ km/h.
>
> **e** Out 30 km, back 30 km: 60 km in total.

C · RATES OF CHANGE AND MISLEADING GRAPHS

5 A plant is 12 cm tall and grows 1.5 cm every week.

a Write a formula for its height h cm after w weeks.

b How tall is it after 8 weeks?

c After how many weeks is it 30 cm tall?

> **a** $h = 12 + 1.5w$. **b** $12 + 12 = 24$ cm. **c** $12 + 1.5w = 30 \Rightarrow 1.5w = 18 \Rightarrow w = 12$ weeks.

6 A shop's poster shows a bar chart of this year's sales (480) next to last year's (450). The vertical axis starts at 440 instead of 0, and the bars have no units. Give two reasons why the chart is misleading.

> Starting the axis at 440 makes a 7% rise look like the bar has quadrupled; without units (or a scale) you cannot tell what the heights mean. (Other fair answers: unequal bar widths, missing title, 3D effects.)

7 Mobile plan A costs €10 a month plus 5 cents per MB of data. Plan B has no monthly fee and costs 10 cents per MB.

- **a** Draw both cost graphs on the same axes for 0 to 300 MB.
- **b** For what amount of data do the plans cost the same?
- **c** Which plan is cheaper for 150 MB, and by how much?

> **a** A: line from $(0, 10)$ with gradient 0.05 — e.g. through $(200, 20)$. B: line from the origin through $(200, 20)$ and $(300, 30)$.
> **b** Where they cross: $10 + 0.05m = 0.10m \Rightarrow 0.05m = 10 \Rightarrow m = 200$ MB (both €20).
> **c** At 150 MB: A Math input error, B Math input error. Plan B, by €2.50.

8 Water freezes at 0 °C $= 32$ °F and boils at 100 °C $= 212$ °F.

- **a** Draw a conversion graph from °C to °F.
- **b** Work out the gradient. What does it tell you?
- **c** Are °C and °F in direct proportion? Explain.

> **a** Straight line from $(0, 32)$ to $(100, 212)$.
> **b** $\dfrac{212 - 32}{100} = 1.8$: each 1 °C rise is a 1.8 °F rise. The formula is $F = 1.8C + 32$.
> **c** No — the line does not pass through the origin (0 °C is 32 °F, not 0 °F).

Check: direct proportion = straight line through the origin · gradient = rate (per hour, per kg, per °C) · on a distance–time graph, steeper means faster and flat means stopped · average speed = distance ÷ time.